

BIMS-LEGAL: Dual-Store Soft Binding for Recovering Prior Legal Advice under Same-Domain Interference

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Abstract

In same-domain legal consultation archives, dense retrieval often surfaces a relevant client question while failing to return the lawyer’s advice turn—an *episode incompleteness* failure that leaves prior recommendations hard to recover under lexical interference from neighbouring matters. BIMS-LEGAL indexes Chinese consultation logs in a dual store: a fast episodic turn index with session identifiers and a slower semantic store consolidated by target- k KMeans. Soft O2 binds session siblings by attenuated score inheritance (βs) rather than hard score copy. Exact replay is a surface-form reference; main channels are paraphrase, follow-up, and advice-recall. On a shared turn-level FlatIP rebuild Soft O2 substantially improves Answer Hit over dense FlatIP; hard hydration is a score-copy envelope that can further raise AH by forcing siblings into top- k . Soft O2-C is evaluated as a conditional cluster-binding mechanism under Mix reconstructions with off-session gold, not as a fully unsupervised deployment claim. We report Answer Hit, Episode Completeness, graded nDCG with fixed gold-session IDCG, a four-way failure taxonomy, and Holm-adjusted Soft O2 vs FlatIP tests.

Keywords: legal information retrieval, consultation memory, soft episode binding, dual-store memory, cluster binding, Chinese legal corpora

1. Introduction

2 A junior associate may ask a firm assistant: “Last month you advised
3 this client on terminating a fixed-term contract—what remedy did we rec-
4 ommend?” The system returns several near-identical “contract termination”
5 questions, ranks the stored user utterance highly, and never surfaces the part-
6 ner’s written advice. The subsequent reply may sound fluent yet be grounded
7 in the wrong episode—or in none. In consultation archives, questions and

8 answers share specialized terminology, neighbouring matters are topically ad-
 9 jacent yet legally distinct, and the professionally useful evidence unit is the
 10 full consultation *episode*.

11 Legal informatics has long studied statutory interpretation, case retrieval,
 12 and advisory systems Martinez-Gil (2023); Ashley (2017), and recent legal
 13 large language models (LLMs) further improve statutory and fact-pattern
 14 answering Yue et al. (2023); Huang et al. (2023); Yue et al. (2024). Classical
 15 legal information retrieval (IR), however, mainly targets *external* authority—
 16 statutes, cases, or community answers—as documents Askari et al. (2024);
 17 Louis et al. (2024); Liu et al. (2022); Zhang et al. (2024). Memory-augmented
 18 dialogue systems recover evidence from evolving conversation logs Wu et al.
 19 (2024); Maharana et al. (2024); Packer et al. (2023), yet firm assistants still
 20 need a reliable index of *internal consultation memory*: prior client-specific
 21 advice buried in a growing same-domain archive under lexical interference
 22 from neighbouring matters.

23 Under that interference, dense turn retrieval often returns a question
 24 turn while its answer sibling remains outside top- k . We term this failure
 25 *episode incompleteness*, as distinct from a full session miss. It is worsened
 26 when approximate indexes distort turn scores and fragment episode context
 27 at moderate archive sizes. Parent-document hydration and session-max ag-
 28 gregation can force siblings into the result list by hard score copy, but they
 29 also erase rank competition among unrelated candidates. What is needed is
 30 a way to complete episodes without collapsing the retrieval ranking into a
 31 session dump.

32 This paper presents BIMS-LEGAL, a dual-store system for Chinese legal
 33 consultation logs. A fast episodic pathway indexes timestamped turn embed-
 34 dings with speaker role and session identifier *sid*; a slower semantic pathway
 35 consolidates turns with target- k KMeans (the production clustering path in
 36 the released implementation). The dual-pathway split is inspired by Comple-
 37 mentary Learning Systems theory McClelland et al. (1995); Kumaran et al.
 38 (2016) only as an architectural analogy. On the episodic pathway, Soft O2
 39 performs session soft binding: when a turn is retrieved with score s , sib-
 40 lings sharing *sid* inherit $\max(s_{t'}, \beta \cdot s)$, so an answer turn can surface under
 41 question-hit / answer-miss while attenuated scores keep candidate competi-
 42 tion intact. On the semantic pathway, Soft O2-C applies the same rule within
 43 KMeans clusters when gold evidence is not co-session with the query; gated
 44 Hybrid is intended to trigger cluster binding only on direct dense hits. Exact
 45 FlatIP indexing (O1) restores faithful turn scores when product quantization

46 collapses Answer Hit; bulk-load consolidation (O3) makes large shared-store
 47 construction tractable without changing retrieval semantics. We evaluate
 48 Soft O2 primarily against FlatIP under episode incompleteness, treat hard
 49 hydration as a score-copy envelope, and examine Soft O2-C under Mix re-
 50 constructions as a conditional mechanism test.

51 The main contributions are as follows. First, we formulate consultation-
 52 memory retrieval under same-domain interference and identify episode in-
 53 completeness as a recurrent failure mode of dense turn retrieval. Second,
 54 we propose Soft O2 as a turn-level soft-binding operator, with a gap-ratio
 55 account of attenuation β and default $\beta=0.98$ chosen by validation sweep,
 56 supported by O1/O3 as implementation operators for shared-store evalua-
 57 tion. Third, we evaluate Soft O2 on CAIL multi-turn and rebuilt LegalEp
 58 corpora against FlatIP, hard hydration, shuffled-*sid*, BM25-turn, RRF, and
 59 cross-encoder controls under a unified FlatIP-rebuild metric pipeline, and re-
 60 port Soft O2-C Mix contrasts as conditional mechanism evidence with graded
 61 nDCG, a four-way failure taxonomy, and Holm-adjusted Soft O2 vs FlatIP
 62 tests.

63 Section 2 reviews related work. Section 3 presents BIMS-LEGAL and
 64 the Soft O2 attenuation analysis. Section 4 details corpora, metrics, and
 65 protocols. Section 5 reports results. Section 6 discusses implications and
 66 limitations, and Section 7 concludes.

67 2. Related Work

68 2.1. Legal IR and long-horizon dialogue memory

69 Legal QA and retrieval systems typically ground generation in statutes,
 70 cases, or published advice Louis et al. (2024); Askari et al. (2024); Martinez-
 71 Gil (2023); Ashley (2017). Conversational legal case retrieval and event-aware
 72 similar-case ranking further show how interaction context and structured le-
 73 gal knowledge shape IR effectiveness Liu et al. (2022); Zhang et al. (2024).
 74 Those pipelines optimize access to external authority. Consultation memory
 75 retrieval targets a different evidence source: *prior consultation episodes* inside
 76 an agent’s own archive. Confusing two clients’ histories can produce fluent
 77 yet professionally inappropriate responses even when statute retrieval is cor-
 78 rect. Domain LLMs such as DISC-LawLLM, Lawyer LLaMA, and LawLLM
 79 improve statutory and advisory generation Yue et al. (2023); Huang et al.
 80 (2023); Yue et al. (2024), but they still need a reliable memory index when
 81 prior advice must be recovered from a growing same-domain log.

Legal consultation archives add a further difficulty: high lexical overlap among topically adjacent matters, with the professionally useful evidence unit being the full question–answer episode. This setting motivates *dual-store* memory architectures that keep fast episodic turn retrieval separate from slower semantic clustering, and session-aware soft binding operators that recover answer turns under same-domain interference without collapsing rank competition among unrelated candidates.

2.2. Dual-store memory and episodic retrieval

Dual-store designs distinguish fast episodic encoding from slower semantic integration McClelland et al. (1995); Kumaran et al. (2016); Tulving et al. (1972); Atkinson and Shiffrin (1971). Memory architectures for dialogue agents extend context via hierarchical paging, summary banks, graphs, and neuro-symbolic retrieval Packer et al. (2023); Zhong et al. (2024); Rasmussen et al. (2025); Gutiérrez et al. (2024); Modarressi et al. (2023); Wang et al. (2024); Lewis et al. (2020); Wu et al. (2025), but most treat retrieved units as flat passages rather than *multi-turn consultation episodes* whose answer turn may rank below an unrelated question turn sharing legal vocabulary. LongMemEval and LoCoMo evaluate long-horizon recall and QA over personal conversation logs Wu et al. (2024); Maharana et al. (2024), while broader memory-agent surveys emphasize persistence beyond the context window Wu et al. (2025); Tan et al. (2025). Recent conversational RAG work stresses reuse of historical search evidence across turns Mo et al. (2026). BIMS-LEGAL positions its dual-store contribution at this intersection: episodic turn embeddings with *sid* for same-session recovery (Soft O2), and KMeans semantic centroids with Soft O2-C for cross-session recovery when query and gold evidence occupy different sessions but share thematic structure. Under dense same-domain interference, a question turn can occupy top- k while its answer sibling is absent. Parent-document hydration and session-max aggregation recover that sibling by hard score copy; Soft O2 instead injects attenuated inherited scores so episode completion does not erase rank competition among unrelated candidates.

2.3. Session-aware retrieval

Parent-document hydration, session aggregation, and joint session indexing are standard session-aware mechanisms for attaching sibling context to a dense hit. Soft episode binding (Soft O2) inherits $\beta \cdot s$ for siblings and keeps ranking competition among candidates. Parent-document and hierarchical

retrieval similarly attach child evidence to a parent unit Callan et al. (1994); Dai et al. (2019); Zaheer et al. (2017); Karpukhin et al. (2020); Soft O2 differs by using attenuated turn-level score propagation rather than hard hydration or atomic session documents. Lexical BM25 Robertson and Zaragoza (2009), dense passage retrieval Karpukhin et al. (2020), dense-sparse Reciprocal Rank Fusion Cormack et al. (2009), and cross-encoder reranking Nogueira et al. (2019); Chen et al. (2024) provide complementary controls outside session membership. Exact flat indexes and product-quantized ANN backends Johnson et al. (2019) further affect whether sibling binding is applied over faithful turn scores.

Contrast with joint-episode indexing. An alternative way to keep episode context is to concatenate full question-answer pairs into monolithic episode documents (joint indexing). Joint indexing can prevent episode incompleteness by construction, but it changes the evidence unit from individual turns to merged documents. That coarser unit weakens fine-grained turn matching—for example, isolating a specific advice sentence—and raises index memory and update cost when new turns arrive in an online consultation log. Soft O2 instead keeps turn-level representations and candidate competition, recovering missing siblings on demand via attenuated score inheritance (βs) without replacing the underlying turn index. Table 1 summarizes the session-structure contrasts used in the main grids.

Table 1: Session-aware mechanisms compared in this work.

Mechanism	Unit	Expansion	Type	Rank
Parent hydration	Session	Insert siblings	Hard	No
Session-max	Session	Session max	Hard	Same*
Joint episode index	Session doc	Atomic	N/A	N/A
Soft O2	Turn+ <i>sid</i>	Soft βs	Soft	Yes
Soft O2-C	Turn+cluster	Soft β_{cs}	Soft	Yes

*On the reported corpora, session-max rankings coincide with hard parent hydration under full-sibling expansion; main tables report a single hard-expansion baseline. Soft O2 denotes session soft binding; Soft O2-C denotes cluster soft binding on the KMeans semantic pathway.

In short, the dual-store split is the system design, and Soft O2 /Soft O2-C are the operators evaluated on Chinese legal consultation archives.

145 3. Method: BIMS-LEGAL

146 3.1. Architecture overview

147 Figure 1 summarizes the pipeline. BIMS-LEGAL maintains complemen-
148 tary episodic and semantic stores over the same consultation archive.

149 **Definition 1** (Episodic and semantic elements). *An episodic turn (episodic
150 element) is $e_i = (\mathbf{v}_i, t_i, c_i, \phi_i, \text{sid}_i)$, where $\mathbf{v}_i \in \mathbb{R}^d$ is an embedding, t_i a times-
151 tamp, c_i conversational context (including speaker role), ϕ_i surface features,
152 and sid_i the consultation session identifier. We reserve session turns for the
153 weaker notion of turns that share only sid under a grouping key, without re-
154 quiring role, timestamp, or other episodic metadata to participate in scoring.
155 Soft O2 binds over session siblings $\text{Sib}(t) = \{t' : \text{sid}_{t'} = \text{sid}_t\} \setminus \{t\}$, but
156 the indexed objects themselves remain episodic turns. A semantic element is
157 $s_j = (\mathbf{c}_j, C_j, \kappa_j, \tau_j)$ with centroid $\mathbf{c}_j$, member set C_j , label κ_j , and formation
158 statistics τ_j . Write $\text{Clus}(t) = \{t' : \text{cid}_{t'} = \text{cid}_t\} \setminus \{t\}$ for semantic-cluster
159 siblings.*

160 **Definition 2** (Dual-store retrieval map). *Given query q with embedding
161 $\mathbf{v}_q$, base episodic retrieval returns a scored candidate set $\mathcal{R}_0(q) = \{(t, s_t)\}$
162 under Eq. (1). Soft O2 produces $\mathcal{R}_{\text{O2}}(q) = \text{SoftBind}(\mathcal{R}_0(q); \text{Sib}, \beta)$ (Sec-
163 tion 3.3). Soft O2-C produces $\mathcal{R}_{\text{O2C}}(q) = \text{SoftBind}(\mathcal{R}_0(q); \text{Clus}, \beta_c)$ (Sec-
164 tion 3.5). Gated Hybrid composes session binding then cross-session cluster
165 binding (Algorithm 2). The returned list is the top- k of the bound set under
166 the rewritten scores.*

167 3.2. Base scoring and implementation operators (O1, O3)

168 For query q with embedding $\mathbf{v}_q$, episodic base retrieval scores a memory
169 turn m by

$$\text{score}(m) = \alpha S(\mathbf{v}_q, \mathbf{v}_m) + (1 - \alpha) R(t_m), \quad (1)$$

170 where S is cosine similarity on ℓ_2 -normalized embeddings and

$$R(t_m) = \max\left(0, 1 - \frac{\Delta_t}{30 \text{ days}}\right), \quad (2)$$

171 with Δ_t the elapsed time between query time and the turn timestamp. Unless
172 stated otherwise, $\alpha=0.7$. Eq. (1) is the ranking score used throughout the
173 legal grids; Soft O2 /Soft O2-C rewrite sibling scores after this fusion step
174 and do not replace Eq. (1).

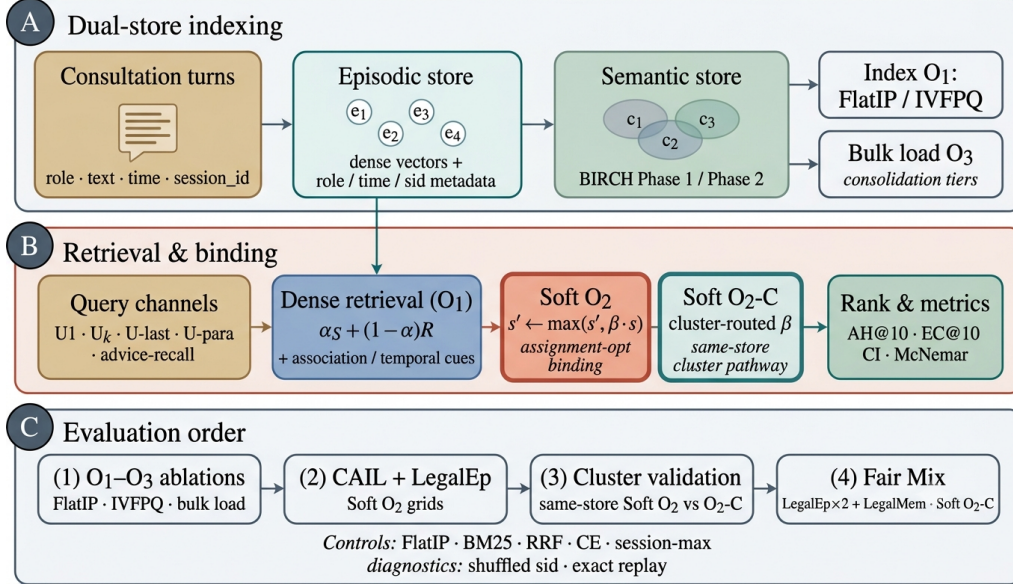


Figure 1: BIMS-LEGAL dual-store pipeline. (A) Episodic turn indexing and KMeans semantic consolidation (O3 bulk-load). (B) Query-time FlatIP retrieval with Soft O2 session binding and Soft O2-C cluster binding; O1 and O3 are implementation operators. (C) Evaluation order: shared-store ablations, Soft O2 grids, and Mix Soft O2-C.

175 *O1: Exact FlatIP indexing..* Let $\hat{S}$ denote the similarity induced by an ANN
 176 backend. At hundreds to a few thousand turns, FAISS IVFPQ is often under-
 177 trained, so $\|\hat{S} - S\|$ is large enough to reorder turns and break Soft O2
 178 inheritance. O1 forces $\hat{S} = S$ via exact FlatIP (inner-product search on
 179 ℓ_2 -normalized embeddings). Quantization error depresses Answer Hit (Ta-
 180 ble A.3: IVFPQ AH 0.330 at $M=1600$ on LegalEp-DISC exact). O1 is an
 181 engineering prerequisite for Soft O2 evaluation, not a standalone retrieval
 182 algorithm.

183 *O3: Bulk-load consolidation..* During bulk construction, O3 replaces the per-
 184 insert update ($e_i \mapsto \text{KMeansConsolidate}(e_i); \text{Flush}$) by a deferred operator
 185 $\text{Finalize} = \text{KMeansConsolidate}(\{e_i\}_{i=1}^N) \circ \text{Flush}$, executed once after all episodic
 186 writes. O3 does not change retrieval semantics; it makes $M \geq 400$ shared-
 187 store experiments runnable (≈ 11 min wall-clock for $M=400$; amortized $\approx$
 188 0.825 s/turn over ~ 800 turns). Quality-latency curves to $M=6400$ appear
 189 in Section 5.6.

190 3.3. Soft O2: session soft binding

191 *Motivation..* Under same-domain interference, a retrieved question turn can
 192 score highly while its answer sibling remains outside top- k —episode incom-
 193 pleteness compounded by IVFPQ score collapse and context fragmentation.
 194 Soft O2 completes episodes without hard score copy.

195 **Definition 3** (Soft binding operator). *Given a scored set $\mathcal{R} = \{(t, s_t)\}$, a*
 196 *sibling relation $\mathcal{N}(\cdot)$, and attenuation $\beta \in (0, 1]$, define*

$$\text{SoftBind}(\mathcal{R}; \mathcal{N}, \beta) : \quad s_{t'} \leftarrow \max(s_{t'}, \beta \cdot s_t) \quad \text{for all } t \in \mathcal{R}, t' \in \mathcal{N}(t). \quad (3)$$

197 *Soft O2 is $\text{SoftBind}(\cdot; \text{Sib}, \beta)$ with default $\beta=0.98$. Hard parent hydration is*
 198 *the special case $\beta=1$ with forced insertion of all siblings; session-max replaces*
 199 *each member score by $\max_{t \in \text{sid}} s_t$.*

200 The released Soft O2 pipeline expands session siblings on raw dense scores,
 201 optionally applies exact-match boost and sparse augment, performs session-
 202 first reranking, fuses semantic and recency scores (Eq. (1)), then runs a final
 203 completeness pass before top- k truncation. Algorithm 1 abstracts the binding
 204 core (expansion + completeness); β therefore acts on the score field used at
 205 each expansion stage as implemented in `memory_manager.py`.

206 3.4. Soft O2 attenuation and score competition

207 The attenuation β controls how strongly a retrieved turn promotes its
 208 session siblings. Availability of the injected sibling rises with β , while fidelity
 209 of the direct hit falls as $\beta \rightarrow 1$. Table 2 summarizes the empirical gap-ratio
 210 distribution $\rho = s_d/s_h$ on LegalEp stores under an exact-channel query proxy:
 211 median ρ lies well below one, and a high- β operating point near 0.98 covers
 212 most measured distractors while remaining strictly below hard hydration
 213 ($\beta=1$). We therefore characterize the availability–ranking trade-off with a
 214 three-candidate score-competition model and choose β by validation sweep
 215 rather than by a closed-form optimum. A zero-margin hinge objective on
 216 $\beta \in (0, 1]$ is uninformative for this choice: its rank term vanishes for all
 217 $\beta < 1$.

218 **Definition 4** (Three-candidate score-competition model). *For a query whose*
 219 *dense retrieval directly hits a gold turn h with fused score $s_h > 0$, let a be*
 220 *the gold answer sibling and let d be the strongest non-session distractor in a*
 221 *wide candidate pool, with scores s_a and s_d . Soft O2 rewrites*

$$s'_h = s_h, \quad s'_a = \max(s_a, \beta s_h), \quad s'_d = s_d, \quad (4)$$

Algorithm 1 Soft O2 session soft binding

Require: Query q , base hits $\mathcal{R}_0$, top- k , attenuation β

```
1:  $\mathcal{R} \leftarrow \mathcal{R}_0$ 
2: for each hit  $t \in \mathcal{R}_0$  with score  $s_t$  do
3:   for each sibling  $t' \in \text{Sib}(t)$  absent from  $\mathcal{R}$  do
4:     insert  $(t', \beta \cdot s_t)$  into  $\mathcal{R}$ 
5:   end for
6: end for
7:                                      $\triangleright$  Completeness pass after fusion with Eq. (1)
8: for each session  $sid$  represented in  $\mathcal{R}$  do
9:    $s_{\max} \leftarrow \max\{s_u : u \in \mathcal{R}, sid_u = sid\}$ 
10:  for each  $t' \in \text{Sib}$  of  $sid$  still absent do
11:    insert  $(t', \beta \cdot s_{\max})$  into  $\mathcal{R}$ 
12:  end for
13: end for
14: return top- $k$  of  $\mathcal{R}$  by descending score
```

Table 2: Gap-ratio statistics on LegalEp $M=3000$ stores (exact-channel query proxy; strongest measured non-session distractor). Default β is chosen by the AH sweep in Table A.4.

Corpus	ρ_{50}	ρ_{95}	Cover@0.90	Default β
LegalEp-DISC	0.746	0.854	0.981	0.98
LegalEp-Lawyer	0.662	0.815	0.998	0.98

222 and defines the gap ratio $\rho = s_d/s_h$. Because d is the strongest measured
223 distractor, conditions based on ρ are conservative sufficient conditions for
224 beating that distractor, not exact top- k boundary conditions for every com-
225 petitor.

226 The binding analysis is most relevant in the soft-injection regime $s_a \leq \beta s_h$,
227 where $s'_a = \beta s_h$.

228 **Proposition 1** (Feasible attenuation band). *Under Definition 4, assume*
229 *the soft-injection case $s_a \leq \beta s_h$. Then (i) availability against the measured*
230 *distractor requires $\beta > \rho$; (ii) rank preservation of the direct hit over the*
231 *injected sibling requires $\beta < 1$, and more generally $s'_a < s_h$; (iii) when $\rho < 1$,*
232 *any $\beta \in (\rho, 1)$ satisfies both conditions in the active injection case. If $s_a > s_h$*

233 *already, rank preservation depends on the observed sibling score rather than*
 234 *on β alone.*

235 *Proof.* If $s_a \leq \beta s_h$, then $s'_a = \beta s_h$. Availability follows from $\beta s_h > s_d$, i.e.
 236 $\beta > \rho$. Rank preservation requires $s'_a < s_h$, which reduces to $\beta < 1$ when
 237 $s_h > 0$. $\square$

238 *Soft-rank visualization and sweep..* On a validation grid we also plot the soft
 239 pairwise surrogate

$$\mathcal{L}_{\text{sr}}(\beta; \rho) = -\log \sigma(\tau(\beta - \rho)) - \gamma \log \sigma(\tau(1 - \beta)), \quad (5)$$

240 with logistic σ , temperature τ , and rank weight γ . The curve is a visualization
 241 of the availability–ranking trade-off under the measured ρ pool; it is not used
 242 as a fitted optimum. Default $\beta=0.98$ is taken from the AH sweep in Table A.4,
 243 where $\{0.90, 0.95, 0.98\}$ form a stable high-AH band on CAIL follow-up and
 244 LegalEp paraphrase as well as exact grids. Advice-recall uses the same default
 245 without a separate gap-ratio fit.

246 Fair controls include hard parent hydration, session-max aggregation, and
 247 shuffled *sid*. On two-turn LegalEp episodes, hard hydration and session-max
 248 coincide under full-sibling expansion, so tables report a single hard-expansion
 249 baseline. Concatenating a full question–answer episode into one retrieval unit
 250 (BM25-joint or dense joint-episode documents) changes the evidence unit
 251 from turns to documents; we treat that design as an alternative indexing
 252 choice rather than a Soft O2 turn-binding baseline (Section 4).

253 3.5. Semantic consolidation and Soft O2-C

254 Semantic consolidation in the released system uses target- k KMeans over
 255 turn embeddings (`encode_with_target_k`), producing cluster ids consumed
 256 by Soft O2-C. An optional sklearn Birch fit remains in the codebase only as
 257 an unused ablation helper and is not on the Soft O2-C evaluation path.

258 **Definition 5** (KMeans semantic consolidation). *Given embeddings $\{e_i\}$ and*
 259 *target cluster count k , O3 runs $(\text{labels}, \text{centers}) \leftarrow \text{KMeans}(\{e_i\}; k)$ once at*
 260 *finalize time and assigns each turn t a cluster id cid_t , inducing $\text{Clus}(t)$ in Def-*
 261 *inition 1. Empty clusters are dropped and labels are remapped to a contiguous*
 262 *range.*

Algorithm 2 Gated Hybrid Soft O2 + Soft O2-C

Require: Base hits $\mathcal{R}_0$, budgets (k, B_c) , attenuations (β, β_c)

```
1:  $\mathcal{R} \leftarrow \text{SoftBind}(\mathcal{R}_0; \text{Sib}, \beta)$  ▷ session pathway first
2:  $\mathcal{S}_{\text{cov}} \leftarrow \{sid_t : t \in \text{top-}k(\mathcal{R})\}$ 
3:  $\mathcal{T}_{\text{trig}} \leftarrow \{t \in \mathcal{R}_0\}$  ▷ direct dense hits only
4: for each trigger  $t \in \mathcal{T}_{\text{trig}}$  do
5:    $s_{\text{max}} \leftarrow \max\{s_u : u \in \mathcal{R}, cid_u = cid_t\}$ 
6:   for each  $t' \in \text{Clus}(t)$  with  $sid_{t'} \notin \mathcal{S}_{\text{cov}}$ , up to budget  $B_c$  do
7:      $s_{t'} \leftarrow \max(s_{t'}, \beta_c \cdot s_{\text{max}})$ 
8:   end for
9: end for
10: return top- $k$  of  $\mathcal{R}$ 
```

263 **Definition 6** (Soft O2-C). *Soft O2-C is $\text{SoftBind}(\cdot; \text{Clus}, \beta_c)$ with default*
264 *$\beta_c=0.95 < \beta$, reflecting noisier thematic neighbourhoods than session mem-*
265 *bership. A per-cluster sibling budget B_c caps $|\text{Clus}(t) \cap \text{Injected}|$ so large*
266 *thematic clusters cannot flood top- k .*

267 Gated Hybrid (Algorithm 2) injects only *cross-session* cluster siblings
268 relative to sessions already covered by dense /Soft O2 hits, and triggers
269 cluster binding only on direct dense hits. This prevents thematic confusers
270 from displacing same-session siblings already recovered by Soft O2. Soft O2
271 applies when query and gold share *sid*. Soft O2-C is the cluster-pathway
272 operator when gold evidence lies in another session of the same thematic
273 cluster.

274 4. Corpora and Evaluation Protocol

275 Figure 2 sketches the shared-corpus design used throughout the legal eval-
276 uation. Gold needles and same-domain distractors occupy one store. Each
277 query probes that shared archive under fixed interference conditions. Pri-
278 vately rebuilt haystacks are avoided so that system comparisons isolate the
279 retrieval operator.

280 4.1. Datasets and query channels

281 Multi-turn authenticity is evaluated on CAIL2024 consultation dialogues.
282 Gold needles are authentic preliminary and final dialogues, with conversation
283 turns completed by label turns where applicable. Query channels cover the

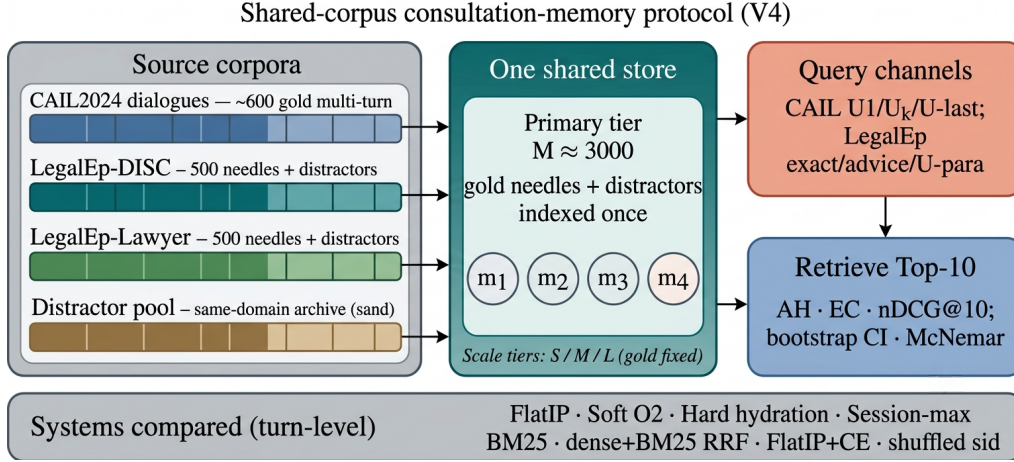


Figure 2: Shared-corpus protocol. Gold needles and same-domain distractors share one store; queries probe multiple channels with turn-level systems and significance tests. Exact replay is diagnostic; paraphrase, follow-up, and advice-recall are primary channels.

284 first user turn ($U1$), a later in-dialogue user turn ($U_k\text{-followup}$), the last user
 285 turn ($U\text{-last}$), and constrained paraphrases ($U\text{-para}$). Evidence defaults to
 286 assistant turns in the target session.

287 LegalEp-DISC and LegalEp-Lawyer rebuild public DISC-Law-SFT and
 288 Lawyer-LLaMA subsets into consultation-episode corpora. Each episode is
 289 a natural (question, answer) pair with session identifier *sid*. Reconstruction
 290 applies length bounds and legal-content filters, excludes exam prompts, re-
 291 moves near-duplicates, separates needles from distractors under a fixed seed,
 292 and retains an audit subsample. After filtering, each LegalEp corpus retains
 293 3500 passed sessions (500 needles, 2500 distractors at the $M=3000$ primary
 294 tier).

295 *Query-channel policy.* Exact replay uses the stored user question and is
 296 reported only as a *diagnostic* upper bound on surface-form recovery. Primary
 297 claims use paraphrase, $U_k\text{-followup}$ / $U\text{-last}$, and advice-recall queries whose
 298 surface form is not identical to the stored question text; paraphrase and
 299 advice-recall templates are generated under a fixed seed and audited on a
 300 subsample. LegalMem-MT reuses CAIL multi-turn gold under the same store
 301 for same-session Soft O2 vs Soft O2-C contrasts and for Mix reconstructions.

302 Archive size is varied with gold needles held fixed. Unless stated otherwise,
 303 Soft O2 transfer tables use the $M \approx 3000$ tier; O1–O2 ablations use $M=400$,

304 $S=300$.

305 4.2. Metrics and baselines

306 Primary metrics are Answer Hit@ k (AH), Episode Completeness@ k (EC),
307 and normalized Discounted Cumulative Gain@ k (nDCG) at $k=10$. AH is
308 binary: whether a gold answer turn appears in the top- k . EC is the mean
309 coverage of gold turns within the target session. Session Hit@ k is reported
310 only for diagnosis. For nDCG we use fixed graded relevance over the full
311 gold session: answer turns $rel=1.0$, other same-session turns $rel=0.5$, all
312 other turns $rel=0.0$. The ideal denominator (IDCG) is computed from all
313 gold-relevant turns in the target session and truncated at k , so every system
314 on a query shares the same IDCG. Answer Hit and Episode Completeness
315 are the main Soft O2 outcomes; nDCG is reported with the fixed gold-session
316 IDCG so that systems remain comparable on each query. Hard expansion
317 can raise nDCG when copied siblings occupy early ranks, so AH and EC are
318 read together with nDCG.

319 The primary hypothesis family is Soft O2 vs FlatIP on the nine CAIL
320 and LegalEp transfer channels; for that family we report McNemar mid- p on
321 paired per-query Answer Hit and Holm-adjusted p -values (Table A.2). Ap-
322 pendix CAIL grids report AH point estimates from the same rebuild; LegalEp
323 appendix grids report AH/EC/nDCG without separate CI columns. Hard-
324 hydration, CE, and Soft O2-C Mix contrasts are secondary.

325 Unless noted, Soft O2 grids in Sections 5.3–5.4 and Appendix Tables A.5–
326 A.7 use one FlatIP rebuild so that AH, EC, nDCG, and failure rates on each
327 row come from the same per-query rankings (`corrected_metrics_*.json`
328 under `paper/ipm/figures/`).

329 Turn-level dense FlatIP (O1) is the base dense retriever. Soft O2 applies
330 soft sibling score inheritance on the episodic pathway; Soft O2-C applies the
331 same rule within KMeans clusters. Hard parent hydration expands a hit
332 to all siblings by copying the trigger score. Lexical controls include BM25
333 over turns and dense+BM25 Reciprocal Rank Fusion (RRF) under the same
334 turn-level evidence unit. BM25-joint and dense joint-episode indexes retrieve
335 pre-concatenated episode documents and therefore answer a coarser retrieval
336 problem; they are discussed as alternative designs in Section 6.3 rather than
337 Soft O2 turn-binding baselines. A cross-encoder (CE) control reranks the
338 top-30 FlatIP candidates with `BAAI/bge-reranker-v2-m3`. Shuffled *sid*,
339 a β sweep, and IVFPQ provide membership, attenuation, and quantization
340 controls.

341 4.3. Fair Mix protocol for Soft O2-C

342 On fully same-session LegalMem grids Soft O2 already recovers gold an-
343 swers that share *sid*, leaving Soft O2-C little exclusive headroom and expos-
344 ing it to thematic confusers. To study off-session recovery we rebuild the *same*
345 LegalEp and LegalMem archives under a Mix protocol: 70% of gold episodes
346 are split into query-side S_a and evidence-side S_b with distinct session identi-
347 fiers, while 30% remain intact. Distractors are unchanged. FlatIP, Soft O2,
348 Soft O2-C, and gated Hybrid share unrestricted dense retrieval on this store;
349 we report overall AH and stratum AH on `cross_session` vs `same_session`
350 queries.

351 *Cluster co-membership and purity under Mix..* After KMeans consolidation,
352 each cross-session S_a/S_b pair is assigned a shared cluster identifier. This
353 construction uses split metadata available only when building the evaluation
354 store; it does not inject labels at query time. The intent is to test Soft O2-
355 C when thematic co-membership is present. Cluster purity still matters: a
356 shared *cid* that also contains many thematically adjacent distractors narrows
357 Soft O2-C’s effective margin, which is why Hybrid gates cluster injection to
358 direct dense hits and caps the per-cluster budget B_c . If recovery still fails
359 under guaranteed S_a/S_b co-membership, the failure is attributable to binding
360 rather than accidental cluster separation. Natural same-cluster rates without
361 this construction are lower (Section 6.3). Soft O2’s session membership rule
362 is unchanged. LegalEp Mix uses advice-recall queries; LegalMem Mix uses
363 mid-split Uk-followup queries. LongMemEval /LoCoMo numbers from prior
364 runs on this codebase (QA correctness 70.7% / 68.2%) supply long-horizon
365 memory context only Wu et al. (2024); Maharana et al. (2024).

366 5. Experiments

367 Unless noted, all dense retrieval grids use the multilingual encoder `nomic-embed-text-v2-moe`
368 Nussbaum et al. (2025) (768-d) as a single shared embedding backend, with
369 $k=10$ and seed 42 on the main $M \approx 3000$ stores. We keep the encoder fixed
370 so Soft O2 vs FlatIP contrasts isolate binding rather than representation
371 changes; sensitivity to alternative Chinese or multilingual encoders (e.g.,
372 `bge-large-zh-v1.5`) is discussed in Section 6.3. A five-seed check on a
373 smaller shared store ($M=400$) is reported later. Primary contrasts appear in
374 Figures 3–5; full numeric grids are in the appendix. Each main table maps
375 to a released JSON artifact in the accompanying repository.

376 5.1. Implementation operators and Soft O2 ablation

377 Table 3 reports the shared-store ablation on DISC-Law and Lawyer-LLaMA
 378 ($M=400$, $S=300$). IVFPQ baseline Answer Hit is 0.540 / 0.397. O1 alone
 379 yields a small gain on DISC (0.567) and a near-tie on Lawyer (0.397), confirm-
 380 ing that exact indexing is a necessary implementation prerequisite but insuffi-
 381 cient alone. O1+Soft O2 raises Answer Hit to 0.780 / 0.680 (+0.240 / +0.283
 382 over baseline) and Episode Completeness to 0.888 / 0.840. O3 is enabled for
 383 all configurations in this ladder as a construction prerequisite; it does not
 384 alter retrieval scores.

Table 3: O1+Soft O2 ablation on the shared store ($M=400$, $S=300$, $k=10$). EC denotes Episode Completeness (gold-turn coverage); AH denotes Answer Hit. O3 bulk-load is on for all rows. Boldface: best AH and EC within each corpus. Graded nDCG for the $M \approx 3000$ grids appears in Table 6.

Corpus	System	EC@10	AH@10
DISC-Law	IVFPQ baseline	0.770	0.540
	+ O1 FlatIP	0.782	0.567
	+ O1+O2 Soft O2	0.888	0.780
Lawyer-LLaMA	IVFPQ baseline	0.698	0.397
	+ O1 FlatIP	0.698	0.397
	+ O1+O2 Soft O2	0.840	0.680

385 5.2. Failure taxonomy

386 We classify each query into four mutually exclusive modes: *complete*
 387 (question and answer in top- k), *incomplete* (question hit / answer miss),
 388 *answer-only* (answer hit / question miss), and *session miss* (no gold-session
 389 turn in top- k). Table 4 reports these rates for FlatIP, Soft O2, hard hydration,
 390 and shuffled *sid* on primary non-exact channels under the unified FlatIP re-
 391 build. Soft O2 converts many incomplete FlatIP cases into complete retrieval
 392 while session-miss stays nearly fixed; shuffled *sid* restores high incomplete-
 393 ness. Hard hydration can drive incompleteness near zero by score copy and
 394 often leads Soft O2 on complete rate; Soft O2 remains the attenuated binder
 395 ($\beta < 1$) rather than a hard session dump (Section 5.4).

396 5.3. Turn-level Soft O2 controls

397 Primary Soft O2 claims use a shared turn-level index. Table 5 reports
 398 Answer Hit for FlatIP, Soft O2, hard hydration, and shuffled *sid* on the pri-
 399 mary Soft O2 transfer channels. Soft O2 beats FlatIP on every non-exact

Table 4: Failure taxonomy ($k=10$) with four mutually exclusive classes from per-query labels: complete (Q+A in top- k), incomplete (Q hit / A miss), answer-only (A hit / Q miss), and session miss. Boldface: best complete / lowest incomplete within each block among FlatIP, Soft O2, and shuffled *sid* (hard hydration shown as a score-copy envelope). Rates sum to 100% within rounding.

Corpus / channel	System	Complete	Incomplete	Ans-only	Session miss
LegalEp-DISC / u-para	FlatIP	70.8%	26.8%	0.8%	1.6%
LegalEp-DISC / u-para	Soft O2	89.1%	8.7%	0.4%	1.8%
LegalEp-DISC / u-para	Hard hydr.	97.0%	0.0%	0.2%	2.8%
LegalEp-DISC / u-para	Shuffled O2	66.0%	31.4%	0.4%	2.2%
LegalEp-DISC / advice	FlatIP	48.0%	17.0%	0.0%	35.0%
LegalEp-DISC / advice	Soft O2	60.8%	3.8%	0.0%	35.4%
LegalEp-DISC / advice	Hard hydr.	64.8%	0.0%	0.2%	35.0%
LegalEp-DISC / advice	Shuffled O2	45.6%	19.2%	0.0%	35.2%
LegalEp-Lawyer / u-para	FlatIP	71.5%	25.3%	2.4%	0.8%
LegalEp-Lawyer / u-para	Soft O2	94.0%	3.8%	0.6%	1.6%
LegalEp-Lawyer / u-para	Hard hydr.	98.0%	0.0%	0.0%	2.0%
LegalEp-Lawyer / u-para	Shuffled O2	70.7%	25.5%	2.2%	1.6%
CAIL / Uk-followup	FlatIP	45.8%	54.0%	0.0%	0.2%
CAIL / Uk-followup	Soft O2	92.8%	7.0%	0.0%	0.2%
CAIL / Uk-followup	Hard hydr.	98.3%	1.5%	0.0%	0.2%
CAIL / Uk-followup	Shuffled O2	34.7%	65.2%	0.0%	0.2%

channel; shuffled *sid* collapses toward or below FlatIP. Hard hydration—full sibling score copy—can exceed Soft O2 AH on the same rebuild (Table 5); Soft O2 is therefore best read as an attenuated binder that recovers missing answers without the unrestricted score-copy envelope of hard expansion (Tables A.5–A.7). BM25-turn is a same-granularity lexical control (Table A.8). Joint episode / document-level indexes change evidence granularity and are scoped as alternative designs rather than Soft O2 turn-binding baselines (Section 6.3).

5.4. Soft O2 on session (CAIL and LegalEp)

Figure 3 and Table 5 summarize AH@10 on authentic CAIL multi-turn channels at $M \approx 3000$ under O1 FlatIP. Soft O2 lifts Answer Hit over FlatIP on every channel under the unified rebuild (U1 0.960 vs 0.832; Uk 0.928 vs 0.458; U-last 0.945 vs 0.425). Shuffled *sid* removes the gain, tying the improvement to correct episode membership. Hard expansion can raise AH

Table 5: Primary Soft O2 controls (AH@10, $M \approx 3000$, unified FlatIP rebuild). Boldface: best AH within each row among FlatIP / Soft O2 / Hard / Shuffled. Soft O2 is compared primarily to FlatIP; hard hydration is the score-copy envelope; shuffled *sid* tests membership dependence.

Corpus / channel	FlatIP	Soft O2	Hard	Shuffled
CAIL / U1	0.832	0.960	0.998	0.695
CAIL / Uk	0.458	0.928	0.983	0.347
CAIL / U-last	0.425	0.945	0.983	0.287
LegalEp-DISC / u-para	0.716	0.895	0.972	0.664
LegalEp-DISC / advice	0.480	0.608	0.650	0.456
LegalEp-Lawyer / u-para	0.739	0.946	0.980	0.729
LegalEp-Lawyer / advice	0.444	0.652	0.674	0.424

CAIL2024 multi-turn results ($M \approx 3000$, turn-level systems)

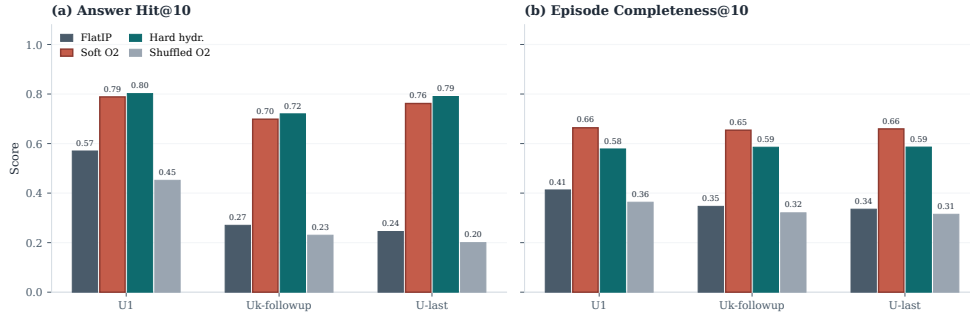


Figure 3: CAIL2024 multi-turn results at tier $M \approx 3000$. Soft O2 improves AH and EC over FlatIP; shuffled *sid* removes the gain; hard expansion can maximize AH at a modest EC cost.

414 further by unrestricted score copy (e.g., Uk 0.983 vs Soft O2 0.928). Soft O2
 415 is therefore the attenuated binder: siblings enter under $\beta < 1$ scores, whereas
 416 hard hydration is the score-copy envelope. Table 6 shows the Soft O2>FlatIP
 417 ordering under graded nDCG with gold-session IDCg, while incompleteness
 418 falls sharply (Table 4). Full numeric grids appear in Tables A.5–A.7.

419 Figure 4 reports LegalEp transfer on the same AH grids. On LegalEp-
 420 DISC exact replay Soft O2 and FlatIP stay near-tied (0.638 vs 0.640), where
 421 both turns are already recoverable for many queries—consistent with treating
 422 exact as diagnostic. On Lawyer exact Soft O2 improves AH to 0.752 from
 423 FlatIP 0.692 (McNemar mid- $p=0.024^*$). Paraphrase yields large Soft O2
 424 gains over FlatIP: 0.895 vs 0.716 on DISC and 0.946 vs 0.739 on Lawyer under

Table 6: Graded nDCG@10 with gold-session IDCG (answer $rel=1.0$, other same-session turns $rel=0.5$) and incompleteness rates on primary non-exact channels. Boldface: Soft O2 better than FlatIP within each channel on nDCG. Soft O2 also exceeds shuffled *sid*.

Corpus	Channel	System	nDCG@10	Incomplete
CAIL	Uk	FlatIP	0.445	54.0%
CAIL	Uk	Soft O2	0.793	7.0%
CAIL	Uk	Shuffled O2	0.369	65.2%
CAIL	U-last	FlatIP	0.420	57.3%
CAIL	U-last	Soft O2	0.791	5.3%
CAIL	U-last	Shuffled O2	0.348	71.2%
LegalEp-DISC	u-para	FlatIP	0.681	26.8%
LegalEp-DISC	u-para	Soft O2	0.759	8.7%
LegalEp-DISC	u-para	Shuffled O2	0.627	31.4%
LegalEp-DISC	advice	FlatIP	0.466	17.0%
LegalEp-DISC	advice	Soft O2	0.524	3.8%
LegalEp-DISC	advice	Shuffled O2	0.436	19.2%
LegalEp-Lawyer	u-para	FlatIP	0.744	25.3%
LegalEp-Lawyer	u-para	Soft O2	0.844	3.8%
LegalEp-Lawyer	u-para	Shuffled O2	0.705	25.5%
LegalEp-Lawyer	advice	FlatIP	0.483	23.4%
LegalEp-Lawyer	advice	Soft O2	0.582	2.2%
LegalEp-Lawyer	advice	Shuffled O2	0.454	25.0%

the unified rebuild. Advice-recall similarly rises to 0.608 from FlatIP 0.480 on DISC and to 0.652 from 0.444 on Lawyer. Hard hydration can still edge Soft O2 by unrestricted score copy on these channels; Soft O2 remains the attenuated alternative ($\beta < 1$). Shuffled *sid* collapses toward FlatIP. Graded nDCG mirrors the Soft O2>FlatIP ordering on paraphrase and advice-recall (Table 6). Detailed grids are in Tables A.6–A.7.

Cross-encoder reranking is a strong neural control at the same turn-level evidence unit: FlatIP+CE reranks the top-30 FlatIP candidates with **bge-reranker-v2-m3**. Table 7 shows that FlatIP+CE is competitive—and on LegalEp-DISC exact, strongest—when surface-form overlap already places both question and answer near the top of the dense pool (DISC exact AH 0.722 for FlatIP+CE vs 0.638 for Soft O2). On channels with semantic shift or multi-turn drift, Soft O2 pulls ahead: CAIL Uk 0.698 vs FlatIP+CE 0.375, CAIL U-last 0.762 vs 0.342, and LegalEp-Lawyer exact 0.752 vs 0.708. A structural limit of the reranker explains the split: CE can refine ranking among candidates already retrieved by FlatIP, but it cannot promote an

LegalEp ($M = 3000$): Soft O2 gains are strongest on paraphrase and remain positive on advice-recall

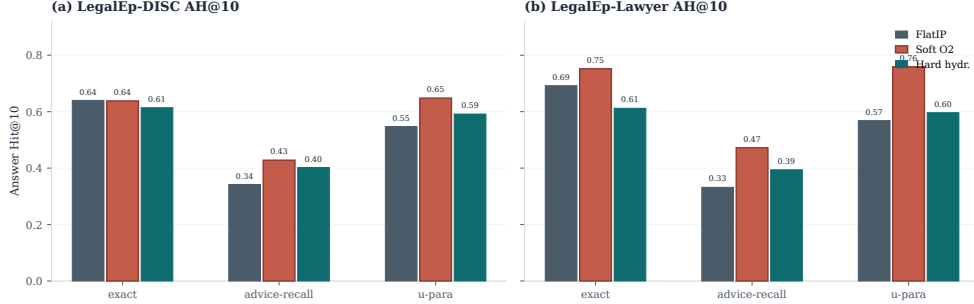


Figure 4: LegalEp AH@10 ($M=3000$, 500 needles). Soft O2 gains are strongest on paraphrase and remain positive on advice-recall; exact transfer is corpus-dependent and treated as diagnostic.

Table 7: Cross-encoder control (AH@10). FlatIP+CE reranks the top-30 FlatIP candidates with `bge-reranker-v2-m3`. Boldface: best AH within each row.

Corpus / channel	FlatIP	FlatIP+CE	Soft O2
CAIL / U1	0.585	0.688	0.788
CAIL / Uk	0.265	0.375	0.698
CAIL / U-last	0.228	0.342	0.762
LegalEp-DISC / exact	0.572	0.722	0.638
LegalEp-Lawyer / exact	0.636	0.708	0.752

441 answer sibling that never entered the top-30 pool. Soft O2 recovers such
 442 siblings by propagating attenuated scores through session links, which helps
 443 precisely when the stored advice turn has weak direct lexical overlap with
 444 the query (paraphrase and advice-recall; cf. Tables 5 and 6).

445 5.5. Soft O2-C and gated Hybrid under Mix

446 When gold answers already share *sid* with the query, Soft O2 is sufficient
 447 and cluster binding is unnecessary. Table 8 asks the off-session question
 448 under the Mix protocol of Section 4.3: 70% cross-session Sa/Sb gold and
 449 30% intact same-session gold, with FlatIP, Soft O2, Soft O2-C, and gated
 450 Hybrid sharing unrestricted dense retrieval.

451 On the cross-session stratum Soft O2-C improves over Soft O2 on LegalEp-
 452 Lawyer (0.491 vs 0.446) and LegalMem-MT (0.537 vs 0.503): cluster binding
 453 can surface split-episode gold that session membership alone misses. Cross-
 454 session Soft O2-C gains motivate gated Hybrid as a selective operator, but

the Mix Hybrid numbers in Table 8 predate a gate fix that records direct dense hits *before* Soft O2 session expansion. Those Hybrid rows are retained only as historical exploratory measurements and are not used for significance claims in this submission; Soft O2-C AH_{cross} contrasts remain the mechanism evidence. Ungated Soft O2-C can raise cross-session hits yet displace intact same-session evidence. Same-session Soft O2 vs Soft O2-C contrasts remain Soft O2-led (Appendix Appendix A, Table A.1).

Table 8: Mix evaluation of Soft O2-C and gated Hybrid (same store, unrestricted dense; AH@10). Gold is 70% cross-session Sa/Sb + 30% intact same-session. Boldface: best overall AH and best cross-session AH within each corpus. Significance vs Soft O2: * $p < 0.05$, ** $p < 0.01$.

Corpus	System	AH	AH _{cross}	AH _{same}	mid- p vs Soft O2
LegalEp-DISC Mix	FlatIP	0.478	0.477	0.480	—
	Soft O2	0.482	0.443	0.573	—
	Soft O2-C	0.410	0.443	0.333	0.0004**
	Hybrid [†]	0.486	0.469	0.527	0.8099
LegalEp-Lawyer Mix	FlatIP	0.462	0.454	0.480	—
	Soft O2	0.496	0.446	0.613	—
	Soft O2-C	0.464	0.491	0.400	0.0857
	Hybrid [†]	0.534	0.523	0.560	0.0105
LegalMem-MT Mix	FlatIP	0.534	0.526	0.553	—
	Soft O2	0.618	0.503	0.887	—
	Soft O2-C	0.500	0.537	0.413	0.0000**
	Hybrid [†]	0.646	0.551	0.867	0.1837

Soft O2-C AH_{cross} is the mechanism contrast. [†]Hybrid rows are exploratory measurements from a pre-fix gate implementation and are not used for primary claims. Cluster co-membership of Sa/Sb pairs follows Section 4.3.

5.6. Scale behaviour

Figure 5 and Table A.3 report quality and warm-query latency on LegalEp-DISC exact ($Q=100$, three repeats; $M \in \{100, 400, 1600, 6400\}$). Soft O2 retains an AH advantage at each tier; IVFPQ collapses at $M=1600$ (AH 0.330), motivating O1; absolute latency grows with M . We treat O1 FlatIP as an engineering prerequisite for Soft O2 at these scales and avoid claiming deployment readiness beyond the measured curve.

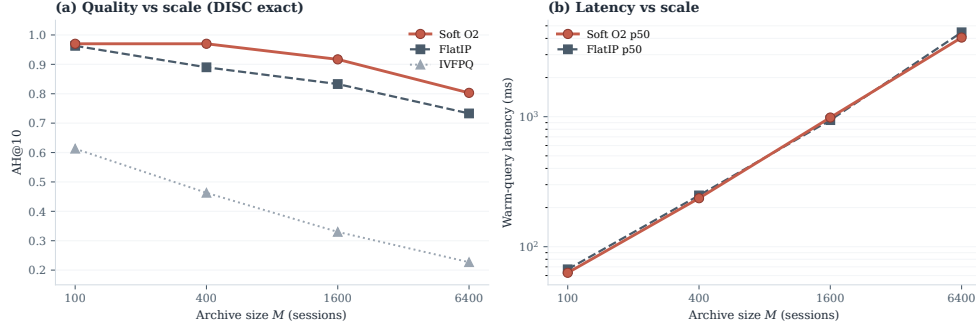


Figure 5: Scale curve on DISC-Law exact. Soft O2 keeps an AH edge over FlatIP as M grows; warm-query latency scales with archive size; IVFPQ quality collapses at moderate M .

472 *End-to-end QA audit (summary)*.. A dual-protocol QA audit ($N=270$ per
 473 corpus; generator `qwen3:14b`, independent judge `qwen3:32b`) reports judged
 474 answerability under Soft O2 retrieval alone; it is not a paired FlatIP vs
 475 Soft O2 generation experiment. Full protocol and stratified results appear in
 476 Section 6.2. On a smaller shared store ($M=400$, five seeds), Soft O2 AH on
 477 LegalEp-DISC is 0.851 ± 0.008 versus FlatIP 0.698 ± 0.040 .

478 6. Discussion

479 6.1. Findings and scope

480 Results show that episode incompleteness is a substantial FlatIP failure
 481 mode under same-domain interference, and that Soft O2 converts many in-
 482 complete retrievals into complete ones without shrinking session-miss rates
 483 (Table 4). O1 restores score fidelity; Soft O2 supplies the main gain through
 484 soft sibling inheritance; O3 makes large shared stores tractable. The gap-
 485 ratio analysis (Section 3.4) motivates keeping β close to but below unity;
 486 the AH sweep peaks in $\{0.9, 0.95, 0.98\}$ rather than at hard hydration ($\beta=1$).
 487 On CAIL under the unified rebuild, Soft O2 raises Answer Hit by large mar-
 488 gins over FlatIP (Uk 0.928 vs 0.458; U-last 0.945 vs 0.425) with collapsing
 489 shuffled-*sid* controls. Hard hydration can exceed Soft O2 AH/EC as a score-
 490 copy envelope. After Holm adjustment on the Soft O2 vs FlatIP family, the
 491 non-exact CAIL and LegalEp gains remain significant, while LegalEp exact
 492 replay is mixed (Table A.2). LegalEp paraphrase and advice-recall, rather
 493 than exact replay, support the main transfer result. Graded nDCG agrees

with the Soft O2>FlatIP ordering (Table 6). Under Mix, Soft O2-C can raise cross-session Answer Hit when thematic co-membership is present (Table 8); Soft O2 remains preferable when gold already shares *sid* (Table A.1). Turn-level BM25, RRF, and CE place Soft O2 among standard IR alternatives Robertson and Zaragoza (2009); Cormack et al. (2009); Nogueira et al. (2019); joint-episode indexes are a coarser design choice outside this protocol.

6.2. RAG application perspective

Consultation-memory retrieval is judged downstream by whether an LLM can answer from recovered evidence rather than confabulate. We audit this end-to-end with a dual-protocol pipeline on Soft O2 retrieval only: generator **qwen3:14b** and independent judge **qwen3:32b** are distinct models. Per corpus we pool $N=270$ judged instances as P1 $n=120$ plus P2 $n=150$ (folder **legal_qa_n270**). The independent judge assigns a continuous evidence-answerability score in $[0, 1]$ from the retrieved context and the generated reply; Evidence-Answerability Rate (EAR) is the fraction of instances with judge score ≥ 0.7 . Table 9 reports pooled EAR and the mean judge score on a stratified subsample ($n=90$ /corpus; 30 each of exact, paraphrase, and follow-up), labelled “Stratified judge” in the table. In the complete-episode stratum—both question and answer turns in top- k —judged answerability is higher in this single-system audit (pooled EAR 0.867 on DISC and 0.859 on Lawyer; Wilson 95% CI $[0.83, 0.90]$ / $[0.82, 0.89]$). Follow-up queries remain the hardest stratum (independent-judge EAR 0.517 / 0.480), consistent with episodic incompleteness persisting even when surface-form overlap is low.

A blind two-annotator legal audit ($n=60$ /corpus) separates retrieval failures (gold episode absent from top- k) from generation failures given retrieved evidence. Annotators used a binary answerability label on each instance; agreement is Cohen’s κ computed on those paired labels (0.71 on DISC, 0.68 on Lawyer). Retrieval-failure share is 0.283 / 0.317. Wrong-session contamination rates stay low (0.083 / 0.100), suggesting that complete episode retrieval under Soft O2 is associated with lower rates of a common RAG failure mode in which the model answers from a neighbouring client’s matter. Hallucinated-legal-basis rates (0.117 / 0.133) remain non-zero, indicating that grounding in retrieved episodes does not guarantee statutory correctness—a scope boundary for deployment. AH and EAR measure evidence availability and judged answerability respectively; statutory correctness, citation validity, and professional liability lie outside this evaluation.

Table 9: End-to-end QA audit ($N=270$ per corpus; generator `qwen3:14b`, independent judge `qwen3:32b`). Pooled EAR: fraction with judge score ≥ 0.7 . Stratified judge: mean judge score on $n=90$ /corpus (exact/paraphrase/follow-up). Human κ : Cohen’s κ on binary answerability labels from two blind annotators ($n=60$ /corpus).

Corpus	Pooled EAR	Stratified judge	Human κ
DISC-Law	0.867	0.757	0.71
Lawyer-LLaMA	0.859	0.729	0.68

530 6.3. Limitations

531 LegalEp episodes are consultation pairs; multi-turn authenticity remains
532 reserved for CAIL.

533 *Dependency on session segmentation and cluster quality.* Soft O2 relies on
534 correct session identifiers (*sid*). In uncured client logs, boundary noise—
535 mid-session topic shifts, multi-intent interruptions, or automated segmenta-
536 tion drift—can corrupt sibling sets. If unrelated turns share a *sid*, attenuated
537 inheritance may propagate false context; if a true advice turn is split into an-
538 other session, Soft O2 cannot bind it. Soft O2-C likewise depends on KMeans
539 cluster purity. Under the Mix protocol, Sa/Sb pairs receive a shared cluster
540 id so binding can be tested when thematic co-membership is present (Sec-
541 tion 4.3); in fully unsupervised consolidation, noisier boundaries shrink the
542 margin over dense retrieval. Deployments on raw consultation streams there-
543 fore need reliable upstream session segmentation, and Soft O2-C should be
544 treated as gated rather than always-on.

545 *Joint-episode indexing.* Concatenating full question–answer pairs into
546 episode documents prevents incompleteness by changing the evidence unit
547 from turns to merged documents (Section 2). That design is outside the
548 Soft O2 turn-binding protocol; a matched comparison under online updates,
549 latency, and fine-grained advice isolation is left to future work.

550 *Embedding model and other scope limits.* All main grids use
551 `nomie-embed-text-v2-moe`. A Chinese-specialized encoder such as
552 `bge-large-zh-v1.5` could alter absolute dense ranks and the measured ρ
553 distribution, and might shrink or widen Soft O2’s relative gain wherever
554 the bi-encoder already places answer turns inside top- k . We did not re-run
555 the full Soft O2 grids under alternative encoders; the binding operator
556 itself is encoder-agnostic once fused scores are available. The ρ statistics

557 use an exact-channel query proxy and the strongest measured distractor;
 558 paraphrase and advice-recall gap laws may shift the feasible band slightly,
 559 though the AH sweep keeps a stable high- β region on those channels. Main
 560 $M \approx 3000$ tables use a fixed seed; five-seed variance is reported only for the
 561 smaller $M=400$ store. LLM paraphrases need human audit. Statutory
 562 correctness, citation validity, and professional liability remain unevaluated.
 563 Because many channels are tested, Holm-adjusted p -values accompany the
 564 Soft O2 vs FlatIP family (Table A.2); Mix Hybrid rows remain exploratory
 565 pending a post-gate re-run and are not used for significance claims.

566 *Latency and deployment scope.* Scale curves cover $M \in$
 567 $\{100, 400, 1600, 6400\}$ under a fixed hardware stack (Table A.3). At
 568 $M=6400$, warm-query p50 latency exceeds 4s (4061 ms for Soft O2; 4458 ms
 569 for FlatIP)—unacceptable for millisecond interactive search. BIMS-LEGAL
 570 should therefore be positioned as **high-precision offline or quasi-**
 571 **real-time consultation memory recovery**—e.g., pre-meeting case
 572 review, audit replay, or batch RAG over a firm’s advice archive—not as
 573 a sub-millisecond search engine. Future work will combine approximate
 574 nearest-neighbour indexes (e.g., HNSW) with Soft O2 and Soft O2-C
 575 binding layered on top of faithful candidate retrieval, preserving episode
 576 completion while reducing query latency. Behavior at roughly 10^5 sessions is
 577 not measured, and we do not claim asymptotic scalability from O3 wall-clock
 578 alone.

579 7. Conclusion

580 BIMS-LEGAL recovers prior advice from Chinese legal consultation
 581 archives under same-domain interference with a dual store: an episodic turn
 582 index and a KMeans semantic store. Soft O2 is the main algorithmic con-
 583 tribution: attenuated session binding completes episodes without hard score
 584 copy, while hard hydration remains a stronger score-copy envelope on the
 585 unified rebuild grids. Soft O2-C extends the same idea to thematic clusters
 586 under Mix reconstructions as a conditional mechanism test. Attenuation β
 587 is chosen by validation sweep under a three-candidate score-competition ac-
 588 count; O1 FlatIP and O3 bulk-load are the implementation operators that
 589 make the shared-store ablations feasible. On CAIL multi-turn and LegalEp
 590 paraphrase /advice-recall channels, Soft O2 improves Answer Hit over FlatIP,
 591 keeps higher episode completeness than hard expansion on CAIL, and raises

592 graded nDCG while cutting incompleteness; shuffled-*sid* controls collapse
593 the gains. Under Mix, Soft O2-C can recover additional cross-session gold
594 when co-membership is present, while Soft O2 remains preferable on intact
595 same-session gold. An end-to-end Soft O2 QA audit associates complete
596 episode retrieval with judged answerability and low wrong-session contam-
597 ination, without claiming paired generation superiority over FlatIP. Turn-
598 level BM25, RRF, and CE place Soft O2 among standard IR alternatives;
599 joint-episode indexing remains a coarser design outside this protocol.

600 **Declarations**

601 *Funding*

602 Computational resources were provided by the Data Law Laboratory,
603 China University of Political Science and Law.

604 *Conflict of interest*

605 The authors declare no competing interests.

606 *Ethics approval*

607 This study uses publicly released research corpora and excludes confiden-
608 tial client data. The system is a research prototype for consultation-memory
609 retrieval.

610 *Data availability*

611 Code, processed evaluation corpora, and primary result JSON files
612 will be made available upon acceptance. Upstream corpora remain un-
613 der their original licenses: CAIL2024 consultation tracks; Hugging Face
614 ShengbinYue/DISC-Law-SFT and Skepsun/lawyer_llama_data. Processed
615 LegalEp /LegalMem artifacts are derived research releases; redistributors
616 should consult each upstream licence before reuse.

617 *Code availability*

618 The BIMS-LEGAL implementation, evaluation pipelines, and ta-
619 ble/figure scripts will be released with the accepted publication.

620 *Author contributions*

621 Author contributions are withheld for anonymous review.

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736 Appendix A. Numeric grids and controls

737 Same-session Soft O2-C contrasts, significance tests, scale curves, and full
738 Soft O2 numeric grids appear below.

Table A.1: Same-session Soft O2 vs ungated Soft O2-C (AH@10). Soft O2 is preferable when gold already shares *sid* with the query; ungated Soft O2-C can surface thematic confusers. Boldface: better AH within each row. Mix Soft O2-C results appear in Table 8.

Corpus / channel	Soft O2	Soft O2-C
LegalEp-DISC / exact	0.836	0.744
LegalEp-DISC / u-para	0.807	0.730
LegalEp-DISC / advice	0.548	0.490
LegalEp-Lawyer / exact	0.818	0.710
LegalEp-Lawyer / u-para	0.817	0.733
LegalEp-Lawyer / advice	0.546	0.448
LegalMem-MT / U1	0.903	0.712
LegalMem-MT / Uk	0.793	0.385

Table A.2: Primary Soft O2 vs FlatIP family: McNemar mid-*p* on paired Answer Hit from V4 `per_query_ah`, with Holm step-down adjustment ($m=9$). Artifact: `paper/ipm/figures/holm_primary_family.json`.

Corpus / channel	Contrast	ΔAH	mid- <i>p</i>	Holm <i>p</i>
CAIL / Uk	O2 vs FlatIP	+0.428	8.99e-48	7.19e-47
CAIL / U1	O2 vs FlatIP	+0.218	5.19e-18	3.63e-17
CAIL / U-last	O2 vs FlatIP	+0.517	5.51e-65	4.96e-64
LegalEp-DISC / exact	O2 vs FlatIP	-0.002	0.9341	0.9341
LegalEp-Lawyer / exact	O2 vs FlatIP	+0.060	0.02386	0.04773
LegalEp-DISC / u-para	O2 vs FlatIP	+0.101	7.95e-05	3.18e-04
LegalEp-Lawyer / u-para	O2 vs FlatIP	+0.191	2.10e-15	1.26e-14
LegalEp-DISC / advice	O2 vs FlatIP	+0.086	1.25e-04	3.76e-04
LegalEp-Lawyer / advice	O2 vs FlatIP	+0.140	1.18e-10	5.89e-10

Table A.3: Scale curve on DISC-Law (mean over 3 repeats). Latency is warm-query p50/p95. Boldface: best AH@10 within each M block; lowest p50/p95 within each M block.

M	Mode	AH@10	Build (s)	p50 (ms)	p95 (ms)
100	FlatIP	0.963	3.0	67	80
100	IVFPQ	0.613	25.2	40	54
100	Soft O2	0.970	3.2	63	69
400	FlatIP	0.890	26.5	248	305
400	IVFPQ	0.463	89.5	171	204
400	Soft O2	0.970	30.9	236	313
1600	FlatIP	0.833	359.8	941	1278
1600	IVFPQ	0.330	251.4	669	751
1600	Soft O2	0.917	393.7	985	1172
6400	FlatIP	0.733	4722.8	4458	4969
6400	IVFPQ	0.227	1072.1	2182	3021
6400	Soft O2	0.803	5245.7	4061	4503

Table A.4: Soft O2 AH@10 versus β on the primary shared-corpus evaluation. Peak performance lies in $\{0.9, 0.95, 0.98\}$; aggressive attenuation ($\beta=0.5$) weakens binding.

Corpus	0.5	0.7	0.9	0.95	0.98	1.0
CAIL / Uk	0.323	0.537	0.697	0.700	0.698	0.662
LegalEp-Lawyer / exact	0.608	0.654	0.754	0.754	0.752	0.752
LegalEp-Lawyer / u-para	0.631	0.637	0.717	0.745	0.759	0.749
LegalEp-DISC / u-para	0.551	0.551	0.618	0.648	0.648	0.642

Table A.5: CAIL2024 shared-corpus results ($M \approx 3000$). Boldface marks the best AH@10 and best EC@10 within each channel (ties bolded jointly). **: Soft O2 vs FlatIP, McNemar mid- $p < 0.01$. nDCG@10 uses gold-session IDCG.

Channel	System	AH@10	EC@10	nDCG@10	95% CI (AH)
U1	FlatIP	0.832	0.554	0.554	—
	Soft O2	0.960	0.924	0.776	—
	Hard hydr.	0.998	0.997	0.864	—
	Shuffled O2	0.695	0.437	0.411	—
Uk-followup	FlatIP	0.458	0.451	0.445	—
	Soft O2	0.928	0.923	0.793	—
	Hard hydr.	0.983	0.979	0.847	—
	Shuffled O2	0.347	0.367	0.369	—
U-last	FlatIP	0.425	0.433	0.420	—
	Soft O2	0.945	0.919	0.791	—
	Hard hydr.	0.983	0.977	0.854	—
	Shuffled O2	0.287	0.359	0.348	—

739

Soft O2 leads EC; hard hydration can lead AH.

Table A.6: LegalEp-DISC ($M=3000$, 500 needles). Boldface: best AH@10 and best EC@10 within each channel. * $p < 0.05$, ** $p < 0.01$ (Soft O2 vs FlatIP, McNemar mid- p). Hard hydration coincides with session-max aggregation; only hard hydration is shown. nDCG@10 uses gold-session IDCG.

Channel	System	AH@10	EC@10	nDCG@10
exact	FlatIP	0.742	0.871	0.730
	Soft O2	0.924	0.962	0.813
	Hard hydr.	1.000	1.000	0.886
	Shuffled O2	0.682	0.841	0.666
advice-recall	FlatIP	0.480	0.565	0.466
	Soft O2	0.608	0.627	0.524
	Hard hydr.	0.650	0.649	0.572
	Shuffled O2	0.456	0.552	0.436
u-param	FlatIP	0.716	0.846	0.681
	Soft O2	0.895	0.937	0.759
	Hard hydr.	0.972	0.971	0.827
	Shuffled O2	0.664	0.819	0.627

Table A.7: LegalEp-Lawyer ($M=3000$, 500 needles). Boldface: best AH@10 and best EC@10 within each channel. * $p<0.05$, ** $p<0.01$ (Soft O2 vs FlatIP). Hard hydration coincides with session-max aggregation; only hard hydration is shown. nDCG@10 uses gold-session IDCg.

Channel	System	AH@10	EC@10	nDCG@10
exact	FlatIP	0.732	0.866	0.753
	Soft O2	0.976	0.988	0.869
	Hard hydr.	1.000	1.000	0.897
	Shuffled O2	0.708	0.854	0.705
advice-recall	FlatIP	0.444	0.559	0.483
	Soft O2	0.652	0.662	0.582
	Hard hydr.	0.674	0.674	0.601
	Shuffled O2	0.424	0.549	0.454
u-param	FlatIP	0.739	0.853	0.744
	Soft O2	0.946	0.962	0.844
	Hard hydr.	0.980	0.980	0.864
	Shuffled O2	0.729	0.845	0.705

Table A.8: Turn-level BM25 and dense+BM25 RRF (AH@10). Boldface: higher AH within each row among reported turn-level cells. BM25-joint / joint-episode document indexes are omitted: they change evidence granularity and were not run as Soft O2 protocol competitors.

Corpus	Channel	AH@10
LegalEp-DISC	exact (BM25-turn)	0.802
LegalEp-DISC	advice-recall (BM25-turn)	0.382
LegalEp-DISC	exact (dense+BM25 RRF)	0.780
LegalEp-DISC	advice (dense+BM25 RRF)	0.394
CAIL	U1 / Uk / U-last (BM25-turn)	0.748 / 0.482 / 0.413
LegalEp-Lawyer	exact / advice (BM25-turn)	0.784 / 0.460